

Q1:

1. Missing product Standards.

Requirements Document Structure

Java Programming Style

Database Design Standards

Documentation Standards

Version Compatibility Standards

2. Ignored Process Standards.

Design Review Conduct

Change Control Process

Version Release Process

Configuration Management Process

Integration Process

3. Technical Risks

Integration Risk

Database Design Risk

Requirement Inconsistency Risk

Testing Risk

Version Compatibility Risk

Req. inconsistency risk

Q2:

a)Solution

Step 1 — Effort Required

Effort required = $72000/750 = 96$ person-months

Step 2 — Monthly Salary per Developer

Monthly Salary per Developer = $160 \times 55 = \$8,800$

Step 3 — Monthly Team Cost

Monthly Team Cost = $6 \times 8800 = \$52,800$

Step 4 — Total Project Cost

Total Project Cost = $14 \times 52800 = \$739,200$

b)Solution

Function Type	Count	Complexity	Weight	Total
EI (External Inputs)	5	Simple	3	15
EO (External Outputs)	2	Simple	4	8
EQ (External Inquiries)	3	Simple	3	9
ILF (Internal Logical Files)	5	Simple	7	35
EIF (External Interface Files)	1	Simple	5	5
Total UFP				72

(a) Calculate Total Function Points

Unadjusted Function Points (UFP)

$$\text{UFP}=15+8+9+35+5=72$$

Adjusted Function Points (FP)

Given:

$$\text{VAF}=1.17$$

$$\text{FP}=\text{UFP}\times\text{VAF}$$

$$\text{FP}=72\times1.17=84.24$$

Answer (a)

$$\text{FP}=84.24$$

(b) Estimate Project Effort

Effort=FP/Productivity

Effort=84.248=10.53 person-months

(c) Calculate Total Project Development Cost

Monthly Salary:

$4000 \times 30 = 120000$

Project Cost:

$10.53 \times 120000 = 1,263,600$

(d) Compute Risk Impact

Risk Impact= $P \times C$

Risk Impact= $0.40 \times 1263600 = 505440$ **Answer (d)**

Risk Impact=PKR 505,440

(e) Revised Final Project Cost

Final Cost=Project Cost+Risk Impact = $1,263,600 + 505,440 = 1,769,040$

Q3:

Part (a) — Validation (positive) ·

Field	Answer
TC ID	TC-01
Module / Submodule	M1 / M1-A (Student ID entry)
Precondition	System is running; login page is displayed
Input	Student ID: 20210045 (8 digits, all numeric)

Steps	1. Open ID entry screen. 2. Enter 20210045. 3. Click Submit.
Expected output	System accepts the ID and navigates to M1-B (eligibility check). No error message shown.
Type	Validation — positive

Part (b) — Validation (negative) ·

Field	Answer
TC ID	TC-02
Module / Submodule	M1 / M1-A (Student ID entry)
Precondition	System is running; login page is displayed
Input	Student ID: 2021AB45 (contains letters, violates BR-01)
Steps	1. Open ID entry screen. 2. Enter 2021AB45. 3. Click Submit.
Expected output	System rejects the input with an error message. M1-B is never invoked.
Type	Validation — negative

Part (c) — Defect ·

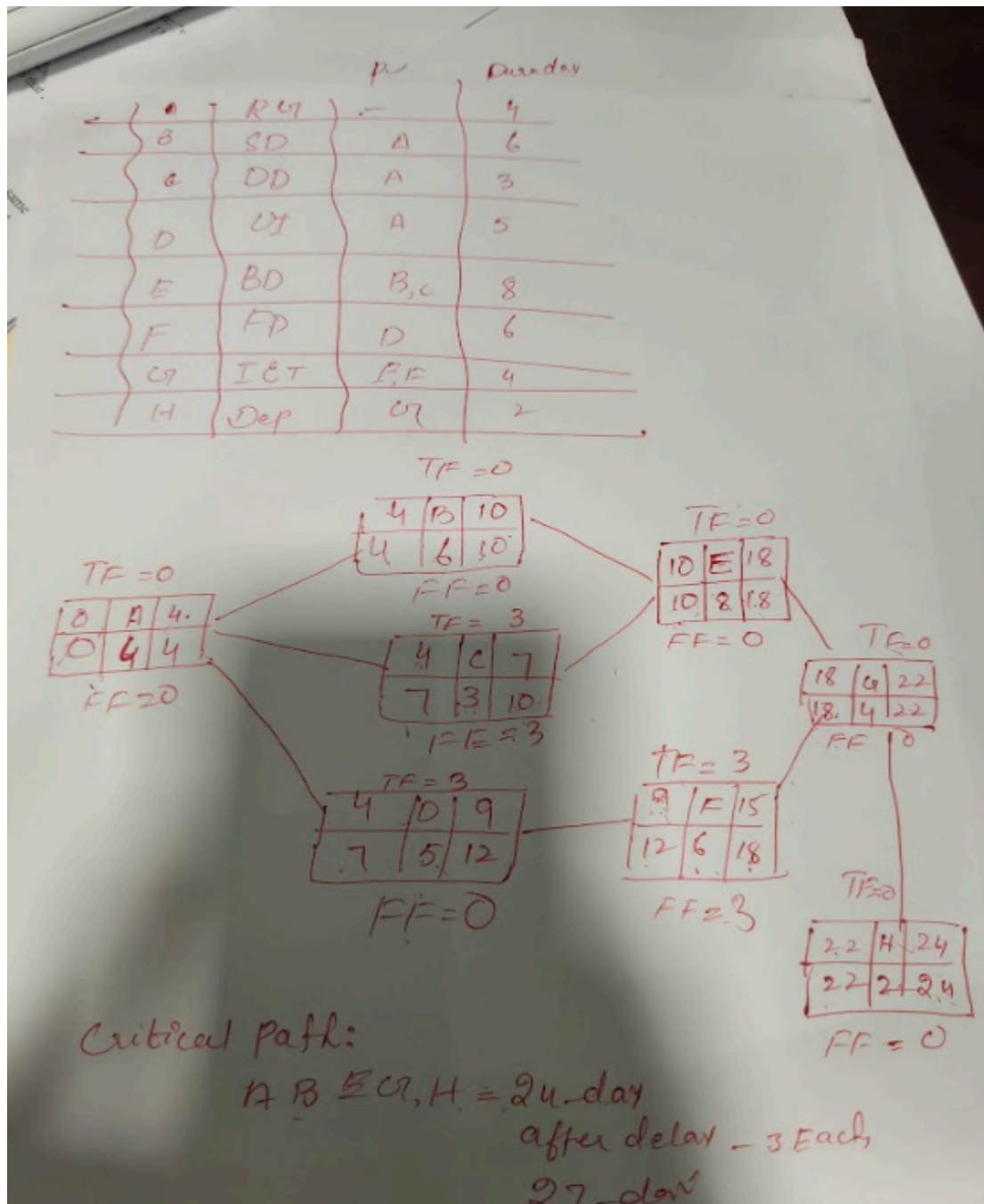
Field	Answer
TC ID	TC-03
Module / Submodule	M2 / M2-B (Issue quota check)
Precondition	Student 20210045 is eligible (<code>fee_status = "paid"</code>); student already has exactly 3 books issued in the database
Input	Request to issue Book ID B-099
Steps	1. Log in as student 20210045. 2. Search for Book B-099 (available). 3. Click Issue.

Expected output	System displays: "Issue limit reached. Return a book first." No new issue record is created.
Type	Defect — boundary condition on BR-03

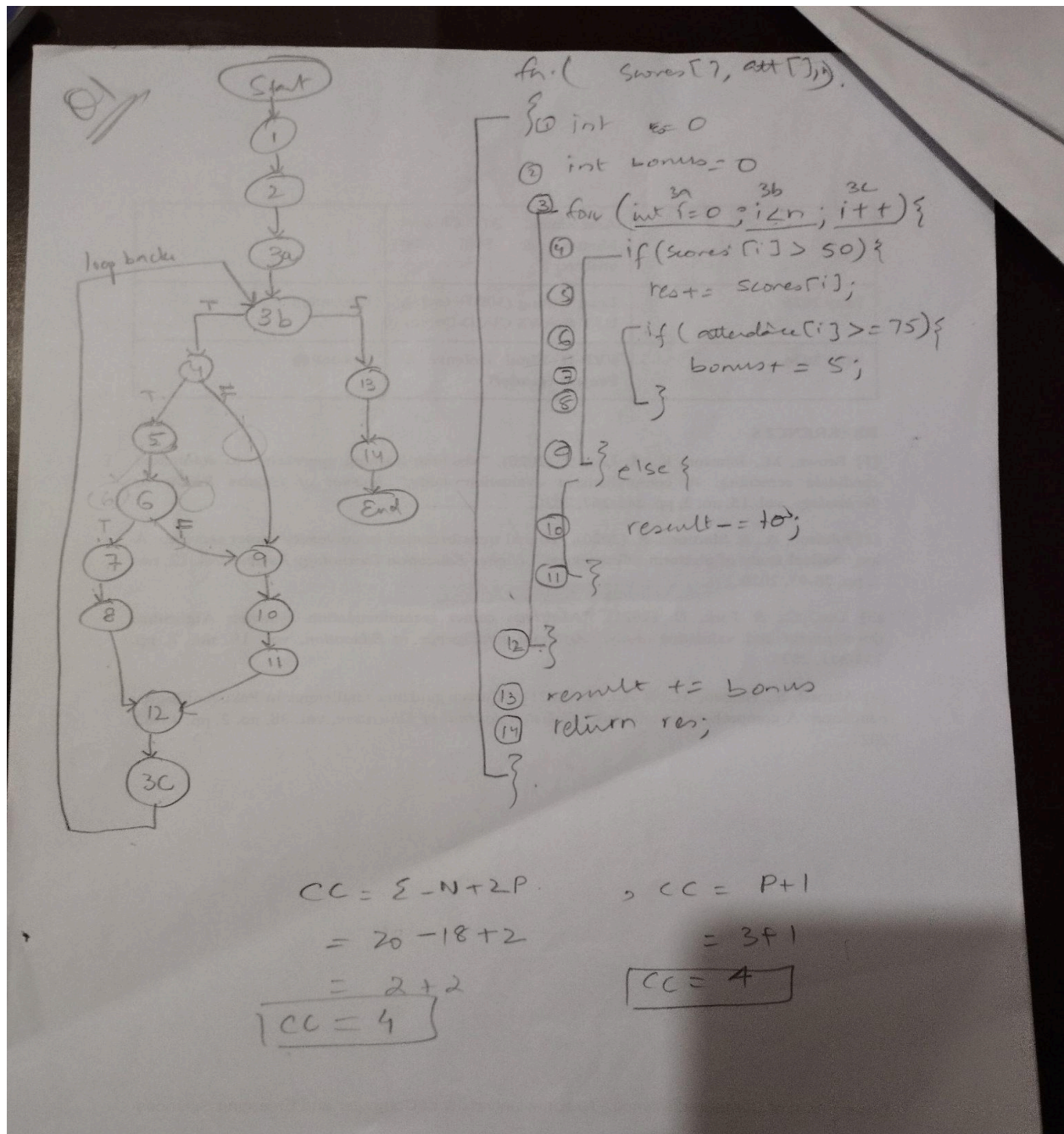
Part (d) — Defect

Field	Answer
TC ID	TC-04
Module / Submodule	M3 / M3-C (Fine amount output)
Precondition	Book B-012 was issued to student 20210045 on 01-May-2026; due date = 14-May-2026; return is processed on 17-May-2026 (3 days overdue)
Input	Return date: 17-May-2026
Steps	1. Open return screen. 2. Enter Book ID B-012. 3. Enter return date 17-May-2026. 4. Click Confirm Return.
Expected output	System displays fine = PKR 15 (3 days × PKR 5). No zero-fine or incorrect amount is accepted.
Type	Defect — fine computation per BR-05

Q4:



Q5:



CC = 4 means there are exactly 4 linearly independent paths. These paths form a basis for all possible execution paths — every other path is a linear combination of these. Therefore, a minimum of **4 test cases** are needed to guarantee that every statement and every branch has been exercised at least once. This gives the tester a quantitative lower bound, which is the core utility of basis path testing.

Q6:

Description:

To handle this project effectively, the Development Team will first break down the modules into smaller user stories and prioritize them according to the Product Owner's priority (Top → High → Medium). Work is then distributed based on team members' skills (backend, frontend, mobile, QA, BA) as follows:

Work Breakdown & Distribution:

- **Backend Developers:** APIs for user management, product catalog, vendor data, authentication , Database Scheme Integration.
- **Frontend Developers:** UI for web modules like user registration, product listing, vendor dashboard, Client site scripting and error handling
- **Mobile Developers:** Mobile app features aligned with web APIs (later priority)
- **Business Analyst (BA):** Refines requirements, writes user stories, clarifies acceptance criteria
- **QA Engineer:** Testing of completed features, writing test cases, bug reporting with all details

Since Modules 1, 2, and 4 are top priority, the team will focus on these first in early sprints.

A) Sample User Stories (Top Priority Modules):

1. User Management Module:

As a new user, I want to register an account using my email and password so that I can access the platform and make purchases.

2. Product Catalog / Listing Module:

As a customer, I want to view and search products by category so that I can easily find items I want to buy.

4. Vendor Dashboard Module:

As a vendor, I want to add and manage my product listings so that I can sell my products on the platform.

b) sprint and sprint goals

Total Sprints: 10 Sprints (6 months = 24 weeks ÷ 2-week sprints = 12 sprints, but using 10 for efficient delivery)

Sprint	Sprint Goal/Milestone
Sprint 1	Module 1 - User Management (Registration, Login, Authentication for Customer/Vendor/Admin)

Sprint	Sprint Goal/Milestone
Sprint 2	Module 2 - Product Catalog (Product listing, search, filter functionality - Backend + Frontend)
Sprint 3	Module 4 - Vendor Dashboard (Product upload, vendor profile, inventory management)
Sprint 4	Module 3 - Shopping Cart (Add to cart, cart operations - Backend + Frontend)
Sprint 5	Module 3 - Checkout & Payment (Payment gateway integration: JazzCash, EasyPaisa, COD)
Sprint 6	Module 5 - Mobile App Phase 1 (Android/iOS - Product browsing, cart, user login)
Sprint 7	Module 5 - Mobile App Phase 2 (Checkout, payments, push notifications for orders)
Sprint 8	Module 6 - Reporting & Analytics (Sales reports, vendor analytics, admin dashboards)
Sprint 9	Integration Testing, Bug Fixes, Performance Optimization, Security Testing
Sprint 10	UAT, Final Testing, Production Deployment, Go-Live across all modules

c)

Challenge A: Product Owner unavailable for a week (Checkout Flow UI completed)

Scrum Master Actions:

1. **Identify Proxy Decision Maker:** Contact PO to temporarily delegate acceptance authority to Business Analyst or senior stakeholder for urgent reviews
2. **Reschedule Sprint Review:** If Sprint Review falls during PO's absence, postpone it by 1-2 days until PO returns
3. **Keep Team Productive:** Guide dev team to move to next priority items in Sprint Backlog while waiting for PO acceptance
4. **Document Completed Work:** Ensure frontend team properly documents the Checkout Flow UI with screenshots for PO's later review

Alignment with Scrum Master Role: SM removes impediments (PO unavailability) and ensures team remains productive without violating Scrum principles (only PO accepts work).

Challenge B: Stakeholders want "Cash on Delivery" payment option mid-Sprint

Scrum Master Actions:

1. **Protect the Sprint:** Remind PO and stakeholders that Sprint Backlog cannot be changed mid-Sprint without team agreement

2. **Facilitate Discussion:** Arrange meeting between PO and dev team to assess if COD can be added without jeopardizing Sprint Goal
3. **Present Options to Team:**
 - If team can accommodate COD by removing equivalent work → Team decides
 - If not feasible → Add to Product Backlog for next Sprint as high priority
4. **Shield Team from Pressure:** Prevent direct stakeholder pressure on dev team; all requests go through PO

Alignment with Scrum Master Role: SM protects team from mid-Sprint disruptions, facilitates collaboration, and ensures Scrum process is followed.

d) A 15-minute daily Scrum meeting where team members briefly share what they completed yesterday, what they will do today, and any blockers they are facing.

Benefits: Improves team communication, ensures transparency, identifies issues early, and keeps everyone focused on the Sprint Goal.